

Data center electricity use and Electric Cars?

Posted by u/Acceptable-Agent-428 • • 0 points • 22 comments

If they are already saying we are having incredibly high energy demand/shortages because of Data Centers, how can we continue to push forward to get more EV's?

The electric companies seem to be saying we can't provide enough power as it is/we are maxed out. So how can they supply the electricity for 10's of thousands of electric cars plugged in as well?

Comments

u/Jgusdaddy • 1 points • 2026-03-16 20:04 UTC

The demand for electric cars is negligible compared to the current demand for ai data centers. It's a non factor. data centers consumed roughly 176–183 TWh of electricity in 2024 (about 4%–4.4% of total U.S. demand), projected to rise to 10%–12% by 2030 due to AI. This demand is rapidly surpassing electric vehicles (EVs), with a single large AI data center consuming as much power as 100,000+ homes, whereas total US EV electricity demand by 2030 is estimated at 2.5%–4.6% of total usage.

Further, electric vehicles can be charged at night, off peak, and use free electricity from backload power that is otherwise wasted. And further, electric vehicles are electrical storage devices, as opposed to data centers and other appliances, which are instant use. You charge at any opportune time, in fact we can programmatically charge at the cheapest time and maybe use them at the time electricity is expensive and finite. Electric vehicle will basically create a golden age economically.

u/Sagrilarus • 2 points • 2025-08-20 12:29 UTC

Solar+battery. It can keep up.

u/[deleted] • 1 points • 2025-08-20 08:46 UTC

[removed]

u/jjllgg22 • 11 points • 2025-08-20 04:19 UTC

Data centers is primarily a resource adequacy and transmission grid congestion challenge. This can is mostly handled by FERC, NERC, and balancing authorities like ISO/RTOs

EVs are primarily a distribution system capacity (and to lesser extent, voltage) challenge. This is handled by state-level regulators. See CPUC "pending load" or NY DSP "coordination grid planning process" to learn how regulators are starting to allow proactive investment in distribution grid in anticipation for EV demand growth

Or see here: https://eta-publications.lbl.gov/sites/default/files/2025-04/8_planning_for_load_growth_transportation_final.pdf

u/disembodied_voice • 15 points • 2025-08-20 03:55 UTC

The "grid can't handle it" talking point usually ignores the fact that [we are building out the grid at a rate well in excess of the anticipated demand increase caused by EVs](<https://cars.usnews.com/cars-trucks/advice/can-the-nations-electrical-grid-support-electric-cars>). Not only that, but the transition to EVs is something that will take decades to complete even in the best case scenario. Even by infrastructural standards, that's plenty of time to adapt.

u/loggywd • 2 points • 2025-08-20 05:29 UTC

China is at 63% BEV for new vehicles sold. If gasoline price becomes higher and EVs get price competitive, the transition can easily happen.

u/Due_Satisfaction2167 • 3 points • 2025-08-20 05:36 UTC

63% NEV.

That includes plug in hybrids and gas range extenders.

u/loggywd • 1 points • 2025-08-20 05:57 UTC

You are right. Anyway it's high number. It's all driven by economics. Current EV sales suck in the US because it's still more expensive.

u/pdx_mom • 1 points • 2025-08-20 04:12 UTC

Transmission and distribution is an issue tho

u/disembodied_voice • 5 points • 2025-08-20 04:17 UTC

Transmission losses [account for only 5% overall](<https://www.eia.gov/tools/faqs/faq.php?id=105>). Even if you account for that, it doesn't change the fact that the grid's capacity is growing faster than the demand growth imposed by EVs.

u/Leonardish • 6 points • 2025-08-20 03:34 UTC

Electricity is a commodity. If we were short on hamburgers, we would build more McDonalds. Same thing. Add capacity, improving transportation and efficiency of distribution and put AI in to manage it all. Humans just add more of what they need.

Here is a great article talking about how China is not only building out massive amounts of new electrical generation, but using it to become the world's first electro-state (as opposed to a petro-state).

https://www.msn.com/en-au/money/markets/china-has-reasons-beyond-the-climate-to-turn-into-the-world-s-first-electrostate/ar-AA1KoWTR

u/jllgg22 • 2 points • 2025-08-20 04:11 UTC

It's very easy to build infrastructure when everything is state-run, like in China

In the US, infrastructure is mostly developed by private enterprise (investor owned utilities), who are beholden to the "cost of service" business model. The US also has a patchwork of permitting agencies that must be navigated, unlike China

All that said, the US has let outdated policies (namely cost allocation), uncoordinated planning processes, and but burdensome permitting get in the way of much-needed bulk system expansion

https://cleanenergygrid.org/wp-content/uploads/2025/07/ACEG_Grid-Strategies_Fewer-New-Miles-2025_Rev-1.pdf

u/zedder1994 • 1 points • 2025-08-20 09:51 UTC

>It's very easy to build infrastructure when everything is state-run, like in China

You obviously know little about China. Plenty of privately run power stations and renewables. They got rid of "everything state run" back in the 80's when Deng was in power.

u/jllgg22 • 1 points • 2025-08-20 12:33 UTC

Wait, so do you believe that China and US have comparable frameworks regarding cost allocation and permitting???

u/Acceptable-Agent-428 • 2 points • 2025-08-20 03:36 UTC

I saw that article. The United States seems to be consistently behind the 8 ball when it comes to almost anything.

u/Mission_Search8991 • 4 points • 2025-08-20 04:06 UTC

Not to get overly political here, but the Far Right has stymied many ideas and initiatives, and they have gotten stronger since Trump 1.0. They now control the tempo and narrative regarding EVs, and anything related to energy in the USA. This will set us back for a long time.

u/jllgg22 • -5 points • 2025-08-20 04:13 UTC

The 3 P's that hold back bulk energy system buildout —price/cost allocation, planning and permitting — have nothing to do with the current administration

But nice try trying to weave in yet another unnecessary political argument into something (I know, in avoidable in any social media)

u/Leonardish • 1 points • 2025-08-24 22:00 UTC

Certainly the US has a number of outdated and inefficient regulatory issues that stymie deployment of the best solutions, but clearly Trump is openly hostile to anything that represents an alternative to fossil fuels. It's like nothing we have ever seen and a serious impediment to transitioning to a lower cost, cleaner energy future.

u/jllgg22 • 1 points • 2025-08-24 22:06 UTC

Yep agreed, last 6 months have been rough for the sector

It was the "anything related to energy in the US" line that was an overreach, as if today's politicians had redesigned our regulatory structure, permitting requirements and grid planning processes which have been holding back the energy transition since it meaningfully began over a decade ago

u/Mission_Search8991 • 8 points • 2025-08-20 04:15 UTC

Have you been asleep the past 6+ months? You may want to do some reading on what is happening.

u/jllgg22 • -5 points • 2025-08-20 04:21 UTC

Have you been asleep the last few decades? Or maybe you're speaking about a topic you know too little about to have a strong position?

Rhetorical question, don't bother answering

u/Mission_Search8991 • 6 points • 2025-08-20 04:25 UTC

Way to gravitate away from Trump and his crowd (drill, baby, drill) stymying EVs, wind, solar, etc, efforts to drag us back

into the 1970s. I think everyone can see where your sentiments lie.

Rhetorical comment, don't bother answering.

u/jjllgg22 • -4 points • 2025-08-20 04:27 UTC

Of course replying with baseless name calling and zero facts. You're clearly very far away from an industry practitioner, so why even bother with the pivot?

Recognize you're speaking out of term and move on. Or keep cramming politics into everything, whatever keeps you happy

This is the primary concern of a doctor. The doctor's job is to help the patient. The doctor is not a person who is interested in the patient's life. The doctor is a person who is interested in the patient's health. The doctor is a person who is interested in the patient's well-being. The doctor is a person who is interested in the patient's happiness. The doctor is a person who is interested in the patient's success. The doctor is a person who is interested in the patient's future. The doctor is a person who is interested in the patient's life. The doctor is a person who is interested in the patient's health. The doctor is a person who is interested in the patient's well-being. The doctor is a person who is interested in the patient's happiness. The doctor is a person who is interested in the patient's success. The doctor is a person who is interested in the patient's future. The doctor is a person who is interested in the patient's life.

